Human pancreatic juice contains zymogens of proteolytic enzymes like trypsinogen, chymotrypsinogen, procarboxypeptidases A and B, pro-elastase along with trypsin inhibitor.

**Trypsin:** Trypsinogen in the pancreatic juice is activated to trypsin by enterokinase and subsequently stabilized by calcium. Trypsin also converts chymotrypsinogen to chymotrypsin, proelastase to elastase and procarboxypeptidases to carboxypeptidases (Figure 2.15). It acts upon the peptide linkages contributed by carboxyl group of Lys and Arg (Figure 2.16). It is ineffective in hydrolyzing the peptide linkages formed by proline. It has an optimum pH of 8-9. Trypsin-inhibitor present in the pancreatic juice prevents the action of trypsin in small intestine and pancreas.

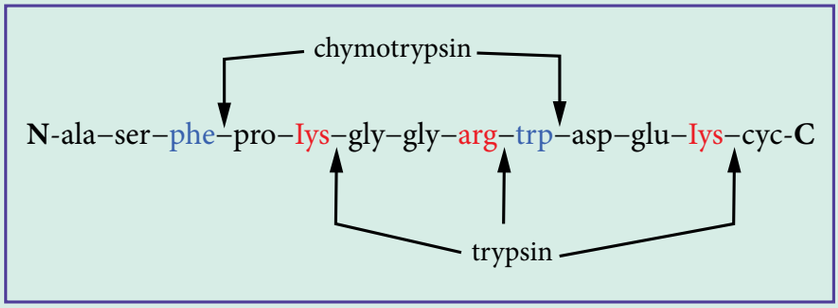



**Figure 2.16 Site of action of chymotrypsin and trypsin**

## Chymotrypsin

Chymotrypsin is also an endopeptidase secreted in the form of its zymogen, chymotrypsinogen, by the pancreas. Its optimum pH is around 8.0. Chymotrypsin catalyzes the hydrolysis of peptide bonds involving the carboxyl group of hydrophobic amino acids like phenylalanine, and tryptophan (Figure 2.17).

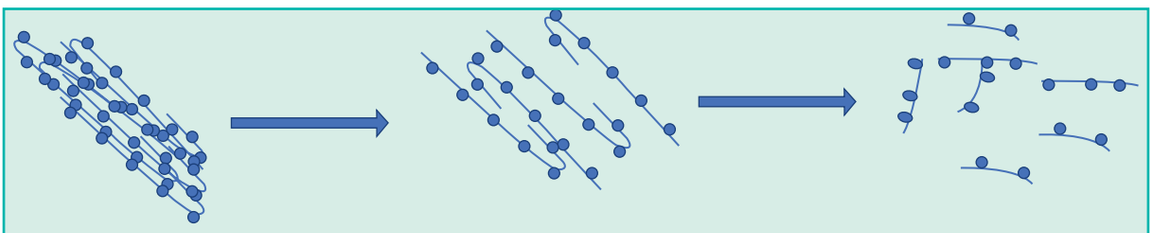


**Figure 2.17 Hydrolysis of peptide bond by chymotrypsin**

## Carboxypeptidases

Carboxypeptidases A and B are zinc containing metallo-enzymes that cleave the peptide bond at the carboxyl or C-terminal end of peptides with a greater preference for aromatic and branched chain amino acids by Carboxypeptidase A and for basic amino acids by Carboxypeptidase B. Elastases and Gelatinases also contribute to the digestion of the proteins. Many dipeptides and tripeptides along with free amino acids are formed in the lumen by the combined action of pancreatic peptidases.



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